# Standard Errors and Confidence Intervals for Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

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Abstract

Mediation modeling using longitudinal data is an exciting field that captures the interrelations in dynamic changes, such as mediated changes, over time. Even though discrete-time vector autoregressive (DT-VAR) approaches are commonly used to estimate indirect effects in longitudinal data, they have known limitations due to the dependency of inferential results on the time intervals between successive occasions and the assumption of regular spacing between measurements. To address these issues, continuous-time vector autoregressive (CT-VAR) models have been proposed as an alternative. Previous work in the area (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) has shown how the direct, indirect, and total effects, for a range of time interval values, can be calculated using parameters estimated from CT-VAR models for causal inferential purposes. However, methods for calculating the uncertainty around the direct, indirect, and total effects in continuous-time mediation have yet to be explored. Drawing from the mediation model literature, we present and compare results from using the delta and MC methods to calculate standard errors and confidence intervals for the direct, indirect, and total effects in continuous-time mediation for inferential purposes. Options to automate these inferential procedures and facilitate interpretations are available in the `cTMed` `R` package.

*Keywords:* mediation, continuous-time, delta method, Monte Carlo method, R package, cTMed

Mediation is a widely adopted model across the social, behavioral, and health sciences. A recent Google Scholar search, conducted as of 8 May 2024, using search terms `"indirect effect OR mediation AND social science OR behavioral science OR health science"`, yielded over five million entries. Even though most mediation work is formulated predominantly as cross-sectional models (Figure 1a), there is a growing trend toward employing longitudinal designs (e.g., Figures 1b, 1c 1d) to address the critical issue of temporal ordering needed to facilitate (causal) interpretations of mediation effects (Cole

& Maxwell, 2003; Georgeson et al., 2023; Maxwell & Cole, 2007; Maxwell et al., 2011; O'Laughlin et al., 2018; Rogosa, 1979). Mediation models help us uncover the intermediate steps through which an independent variable $X$ (such as an intervention) influences a dependent variable $Y$ (such as an outcome) through one or more mediator variables $M$ (e.g., Baron & Kenny, 1986; MacKinnon, 2008). By identifying these mediating variables, we gain insights into the direct and indirect causal pathways that connect interventions to their effects (Fairchild & MacKinnon, 2014; MacKinnon, 1994; Mackinnon & Dwyer, 1993; O'Rourke & MacKinnon, 2018, 2019).

The aim of this manuscript is to provide methods to quantify and visualize the uncertainty associated with point estimates of direct, indirect, and total effects as they change across a range of time interval values in continuous time (CT) using the continuous-time vector autoregressive (CT-VAR) framework. We do this with repeated measurements over time as a stronger basis for causal inference[1] in contrast to cross-sectional designs (Gollob & Reichardt, 1987, 1991). Furthermore, we take the CT framework as previous research (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) have shown its benefits particularly in addressing the time interval dependency problem, that is, direct, indirect, and total effect sizes change as a function of the time interval used to measure the variables. Our novel contributions in this article are to introduce and evaluate use of the delta method (Casella & Berger, 2002; Oehlert, 1992; Ver Hoef, 2012) and the Monte Carlo method (MC) (MacKinnon et al., 2004; Preacher & Selig, 2012) as ways to quantify the uncertainty associated with point estimates of direct, indirect, and total effects across a range of time intervals. Furthermore, we propose using the delta and MC methods to generate regions of significance to help depict and visualize over which range of time interval values the direct, indirect, and total effects are statistically different from zero. Lastly, the methods we developed as well as visualization tools are made available in a free and open-source `R` package `cTMed`.

This manuscript is organized into the following sections: (1) introduction of CT mediation and the need to quantify uncertainty associated with mediation effects; (2) to illustrate, using benchmark examples from the CT mediation literature, the utility of performing inferences on mediation effects in CT using confidence intervals obtained from the delta and MC methods, and regions of significance across a range of time intervals; (3) a Monte Carlo simulation study examining the respective finite-sample performance of these

[1]For observational data, we limit our treatment of causality following the definition provided by Granger (1969), that is, when one variable, denoted as $X$, Granger-causes another variable, denoted as $Y$, it implies that past values of $X$ contain information that helps predict future values of $Y$ beyond what can be predicted using only past values of $Y$ itself.

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confidence intervals; and (4) an application of the proposed inferential approaches to an empirical experience sampling data set studying the indirect effect of physical discomfort on self-esteem through negative affect. We close with a summary of our novel contributions, limitations of the current work, and future directions.

## Mediation Using Longitudinal Data

In this section we discuss the rationale for moving from DT to CT mediation. We provide an overview of the CT modeling framework, namely the linear stochastic differential equation model. While application of the CT framework for mediation models exist in the literature (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021), quantifying the uncertainty associated with mediation effects are yet to be developed. The last part of this section outlines how we address this research gap.

### From Discrete-Time to Continuous-Time Mediation

Mediation and causality have always been linked. From the "causal steps" approach (e.g., Baron & Kenny, 1986; James & Brett, 1984; Judd & Kenny, 1981) to the longitudinal framework (e.g., Cole & Maxwell, 2003; Gollob & Reichardt, 1987, 1991; Maxwell & Cole, 2007; Maxwell et al., 2011), the idea is that causality operates not only directly ($X \rightarrow Y$), but also indirectly ($X \rightarrow M \rightarrow Y$). Gollob and Reichardt (1987, 1991) proposed three principles of causality namely (1) "causes take time to exert their effects and, therefore, causal variables must occur before outcome variables"; (2) "variables can have effects in themselves"; (3) "size of the effect typically varies with the length of time lag".

Principles 1 and 2 underscore the fact that cross-sectional mediation presented in Figure 1a cannot capture the direct and indirect causal associations between $X$, $M$, and $Y$ and the need to move to a model where previous values of variables are used to predict their current values as in the autoregressive cross-lagged mediation model presented in Figures 1c and 1d. The term autoregressive (AR) refers to allowing the variables to have effects in themselves and the term cross-lagged or crossregressive (CR) refers to using previous values of other variables to predict current values of a variable. The umbrella term used for these types of models is discrete-time vector autoregressive model (DT-VAR). The term *vector* autoregressive (VAR) means that there are more than one variable in the system. The term discrete-time (DT) means that we observe the variables at equally spaced time intervals (de Haan-Rietdijk et al., 2017). DT-VAR encompasses panel models where the observations are made in a few time points as well as data from intensive longitudinal data collection where many repeated observations are made.

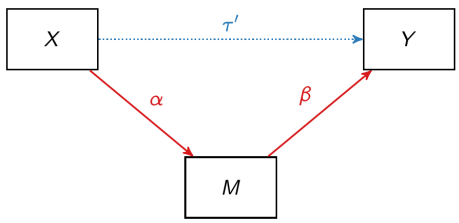
Principle 3 underscores the time interval dependency, that is, in DT-VAR, effect sizes change as a function of the time interval used to measure the variables (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021; Wang & Zhang, 2020). This means that, given the same underlying causal process, the time elapsed (e.g., daily, monthly, yearly) between a "cause" (i.e., the independent variable) and its effect (i.e., the dependent variable) influences the sizes of the effects we observe.

In a cross-sectional mediation, if $X$ changes by one unit, then $Y$ changes by $\tau'$ units, holding $M$ constant. This is the direct effect of $X$ on $Y$. If $X$ changes by one unit, then $M$ changes by $\alpha$ units as a result. Similarly, if $M$ changes by one unit, then $Y$ changes

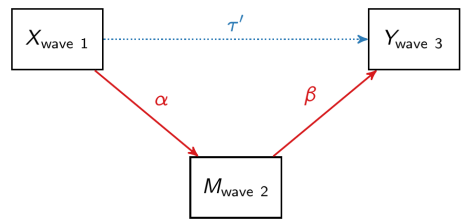
**Figure 1**

*Cross-Sectional and Longitudinal Mediation Models*

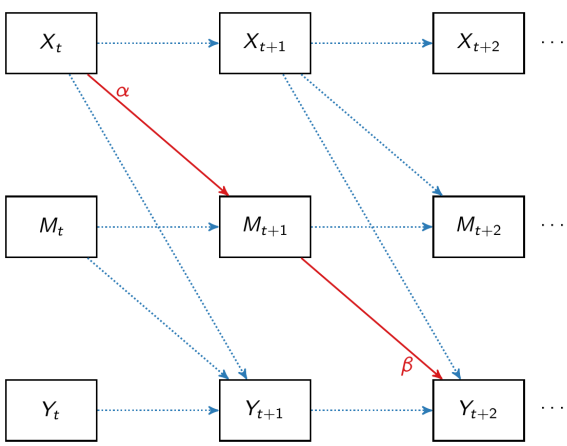
(a) *Cross-sectional mediation model.*



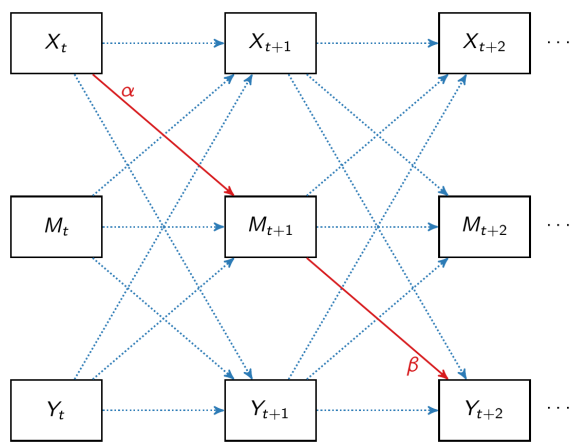

(b) *Three-wave longitudinal mediation model.*




(c) *Unidirectional autoregressive cross-lagged mediation model.*




(d) *Bidirectional autoregressive cross-lagged mediation model.*




*Note:* The indirect effect of $M$ on the path from $X$ to $Y$ is the product of $\alpha$ and $\beta$. For cross-sectional mediation (1a), variables are typically measured on the same time point. The three-wave longitudinal mediation is an improvement to cross-sectional mediation (1b) where variables are measured at three distinct time points. Waves 1, 2, and 3 may or may not be equally spaced. Autoregressive cross-lagged mediation goes further by allowing variables to have effects on themselves. Here $t$, $t+1$, and $t+2$ typically represent equally spaced time points. The dots indicate that the model can accommodate more time points. We distinguish between the unidirectional model (1c), where paths from $M$ to $X$, $Y$ to $M$ and $Y$ to $X$ are constrained to zero and the bidirectional model (1d), where all paths from the previous time point to the current are estimated.

by $\beta$ units as a result. The resulting change of $\alpha$ units on $M$ produced by a unit change in $X$ would then produce a corresponding change of $\beta$ units on $Y$. Thus, $Y$ changes by $\alpha\beta$ units through $M$ as $X$ changes by one unit. This is the indirect effect. The total effect is the sum of direct and indirect effects. In DT-VAR, this idea of decomposing effects also holds. However, we are tracing effects from previous values of the variables to the next. The indirect effect of $X$ at time $t$ on $Y$ at time $t+2$ is the product of the path from $X$ at time $t$ to $M$ at time $t+1$ and the path from $M$ at time $t+1$ to $Y$ at time $t+2$. While many mediation models still rely on cross-sectional data, the good news is that there is a growing trend toward employing DT-VAR models[2]. However, the indirect effects and all the other effects we observe in DT-VAR changes on account of the time elapsed between the observed variables. Thus, temporal precedence, one of the properties that make causal interpretations plausible, also generates new challenges, especially in cross-validating mediation results across studies and designs. A change in modeling framework from DT to CT can address this issue.

The challenges associated with DT-VAR can be addressed by adopting a CT-VAR modeling framework. CT-VAR parameters are defined at any time interval, as such we can obtain the direct, indirect, and total effects for any time interval. From this framework, we can explore how effects change as a function of time interval. Additionally, unlike DT-VAR, which assumes equally spaced data points, CT-VAR accommodates unequally spaced data–a common occurrence in intensive longitudinal studies. These models allow us to capture the evolving dynamics of parameters across a range of time intervals (Boker, 2002; Oud & Delsing, 2010; van Montfort et al., 2018; Voelkle et al., 2012). However, embracing a CT perspective necessitates a conceptual shift in the definition for "total" effect (Aalen et al., 2012, 2016; Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) using CT terminologies, which we review briefly next.

### CT Modeling Using Linear Stochastic Differential Equation (SDE)

In our modeling approach, we consider processes that evolve continuously over time. This means that the variables can assume values at any given moment. To capture this behavior, we employ linear stochastic differential equations (SDEs). Specifically, we focus on a special type of SDE known as the Ornstein–Uhlenbeck model (Chow et al., 2023; Oravecz et al., 2011; Uhlenbeck & Ornstein, 1930).

Similar to structural equation modeling, we can break down our model into two key components: the measurement component and the structural component (Chow et al., 2010). The measurement model describes the relationship between observed variables and latent (unobserved) variables. Specifically, consider the following equation:

$$\mathbf{y}_{i,t_m} = \boldsymbol{\nu} + \boldsymbol{\Lambda}\boldsymbol{\eta}_{i,t_m} + \boldsymbol{\varepsilon}_{i,t_m}, \qquad \boldsymbol{\varepsilon}_{i,t_m} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Theta}) \tag{1}$$

where $\mathbf{y}_{i,t_m}$ represents a vector of observed random variables, $\boldsymbol{\eta}_{i,t_m}$ a vector of latent random variables, and $\boldsymbol{\varepsilon}_{i,t_m}$ a vector of normally distributed random measurement errors, for time $t$ at measurement occasion $m$ and individual $i$. $\boldsymbol{\nu}$ denotes a vector of intercepts, $\boldsymbol{\Lambda}$ a matrix of factor loadings, and $\boldsymbol{\Theta}$ the covariance matrix of $\boldsymbol{\varepsilon}$. In the context of a mediation model

[2]See the decrease in cross-sectional studies from the systematic review by Maxwell and Cole (2007) to O'Laughlin et al. (2018).

with three variables where the effect of the variable $X$ on the variable $Y$ is mediated by the variable $M$, we define $\mathbf{y}$ and $\boldsymbol{\eta}$ as follows:

$$\mathbf{y} = \begin{pmatrix} X \\ M \\ Y \end{pmatrix}, \quad \text{and} \quad \boldsymbol{\eta} = \begin{pmatrix} \eta_X \\ \eta_M \\ \eta_Y \end{pmatrix}. \tag{2}$$

The structural model for $\boldsymbol{\eta}$ in a linear SDE framework focuses on delineating how $\boldsymbol{\eta}$ changes over an infinitely small time interval, $\Delta t$, as $\Delta t$ approaches 0. Using the Ornstein-Uhlenbeck model as the representation for change, for example, the structural model is commonly referred to as the *dynamic* model, and it is given by

$$\mathrm{d}\boldsymbol{\eta}_{i,t} = -\boldsymbol{\Phi}\left(\boldsymbol{\mu} - \boldsymbol{\eta}_{i,t}\right)\mathrm{d}t + \boldsymbol{\Sigma}^{\frac{1}{2}}\mathrm{d}\mathbf{W}_{i,t} \tag{3}$$

In the current example $\boldsymbol{\Phi}$ is the SDE drift matrix with negative diagonals. Appending a negative sign in front makes it $\boldsymbol{\beta}$, where the diagonals are positive. My goal is to have the same matrix for the SDE and OU.

where $\boldsymbol{\mu}$ is the equilibrium or mean value, $\boldsymbol{\Phi}$ is the drift matrix, $\boldsymbol{\Sigma}$ is the matrix of volatility or randomness in the process (also known as the diffusion matrix), and $\mathrm{d}\boldsymbol{W}$ is a Wiener process or Brownian motion, which represents random fluctuations. The Ornstein–Uhlenbeck process centers around a central tendency denoted by $\boldsymbol{\mu}$. When the process deviates from this mean, it naturally gravitates back toward it when the stability condition is satisfied, namely, when all eigenvalues of the matrix $\boldsymbol{\Phi}$ are positive (Oravecz et al., 2011). We can think of $\boldsymbol{\mu}$ as an "attractor" that pulls the process back to its typical value. When we specify the model such that the latent state variables have a stable mean vector $\boldsymbol{\mu}$ consisting of zeros, Equation 3 simplifies to

$$\mathrm{d}\boldsymbol{\eta}_{i,t} = \left(\boldsymbol{\Phi}\boldsymbol{\eta}_{i,t}\right)\mathrm{d}t + \boldsymbol{\Sigma}^{\frac{1}{2}}\mathrm{d}\mathbf{W}_{i,t}. \tag{4}$$

Elements of the matrix $\boldsymbol{\Phi}$ control the speed at which each of the processes reverts to its own equilibrium value, $\boldsymbol{\mu}$. The diagonal elements in $\boldsymbol{\Phi}$, referred to as the auto effects, govern how quickly the variable returns to its mean after being perturbed in the absence of influences from other processes. Smaller absolute values result in slower reversion, allowing the variable to exhibit more persistent behavior away from its mean. The off-diagonal elements of $\boldsymbol{\Phi}$ represent associations between variables and their rates of change. A negative cross effect indicates that if one variable takes on a positive value, there is a resulting decrease in another variable's value. The magnitude of this effect is determined by the absolute value. The diagonal elements of the matrix $\boldsymbol{\Sigma}$ determine the magnitude of volatility around the central tendency $\boldsymbol{\mu}$. Larger values indicate greater fluctuations or noise in the process, while smaller values result in smoother trajectories. Essentially, $\boldsymbol{\Sigma}$ quantifies the degree of randomness or shock-induced variability. The off-diagonal elements of $\boldsymbol{\Sigma}$ represent correlations between random shocks across variables in the system. In summary, the Ornstein–Uhlenbeck process is a mean-reverting stochastic process. Over time, it drifts toward its equilibrium value $\boldsymbol{\mu}$, experiences volatility (as captured by $\boldsymbol{\Sigma}$), and reverts back to the mean with a rate determined by $\boldsymbol{\Phi}$. For the context of this manuscript, we refer to the Ornstein–Uhlenbeck model as the first-order CT-VAR.

The drift matrix $\boldsymbol{\Phi}$ holds particular significance in the context of mediation models. It serves as a conduit for understanding the associations between the states of the variables under investigation. Notably, within a trivariate model, involving $\eta_X$, $\eta_M$, and $\eta_Y$, numerous paths from one variable to another can be explored. This comprehensive connectivity arises because we can construct the model such that all variables are interconnected. Consequently, beyond the classic mediation model $\eta_X \rightarrow \eta_M \rightarrow \eta_Y$, we can also investigate alternative mediation pathways, such as $\eta_Y \rightarrow \eta_M \rightarrow \eta_X$, using the same data set. For our current focus, we will delve into the specific $\eta_X \rightarrow \eta_M \rightarrow \eta_Y$ path. In the subsequent section, we will explore the relationship between CT-VAR and DT-VAR models, emphasizing the pivotal role played by the drift matrix $\boldsymbol{\Phi}$ within the $\eta_X \rightarrow \eta_M \rightarrow \eta_Y$ mediation model.

***Link Between CT-VAR and DT-VAR***

To understand the connection between CT-VAR and DT-VAR models, we need to examine the state space model parameterization of the CT-VAR model which is the DT solution in integral form given by

$$\begin{aligned} \boldsymbol{\eta}_{i,t_{m_i}} &= \boldsymbol{\alpha}_{\Delta t_{m_i}} + \boldsymbol{\beta}_{\Delta t_{m_i}} \boldsymbol{\eta}_{i,t_{m_i-1}} + \boldsymbol{\zeta}_{i,t_{m_i}}, \\ \text{with} \quad \boldsymbol{\zeta}_{i,t_{m_i}} &\sim \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Psi}_{\Delta t_{m_i}}\right) \end{aligned} \tag{5}$$

where $\boldsymbol{\eta}_{i,t_{m_i}}$ and $\boldsymbol{\eta}_{i,t_{m_i-1}}$ represent the vector of latent variables in the current and previous measurement occasion $m$ while $\boldsymbol{\zeta}_{i,t_{m_i}}$ is the normally distributed error term in the current measurement occasion. The model parameter $\boldsymbol{\alpha}$ denotes a vector of intercepts, $\boldsymbol{\beta}$ a matrix of coefficients describing the impact of the lagged values $\boldsymbol{\eta}_{i,t_{m-1}}$ on the current value $\boldsymbol{\eta}_{i,t_m}$ (AR coefficients on the diagonal and CR coefficients on the off-diagonal), and $\boldsymbol{\Psi}$ the covariance matrix of $\boldsymbol{\zeta}$. Notice that the parameters have subscript $\Delta t$, representing that their values are dependent on the time interval, that is, given the same continuous process, their values change depending on the time interval.

In the context of CT-VAR models, the underlying process can take values at any arbitrary time point. However, our observations occur discretely. This means that while the process itself operates continuously, we only observe it at specific discrete time instances. Let's compare this to DT-VAR, where observations are made at specific discrete time intervals (usually positive integers). In a DT-VAR example, suppose we collect data every hour from 7 AM to 10 AM for four measurement occasions. We make discrete observations in time $t$, scaled by the hour, on specific measurement occasions $m = \{1, \cdots, 4\}$. These occasions correspond to time points: $t_1 = 7$, $t_2 = 8$, $t_3 = 9$, and $t_4 = 10$. The time interval from one measurement occasion to the next is constant and equal to 1 hour. In the CT-VAR model, observations can occur at any time between 7 AM and 10 AM for the same four measurement occasions. Specifically, the time points are given by: $t_1 = 7.166667$ (equivalent to 7:10 AM), $t_2 = 8.283333$ (equivalent to 8:17 AM), $t_3 = 9.45$ (equivalent to 9:27 AM), and $t_4 = 10$ (equivalent to 10:00 AM). Notably, the time interval from one measurement occasion to the next varies allows for irregularly spaced observations. We denote this time interval as $\Delta t$, representing the difference between the current measurement occasion $m$ and the previous measurement occasion $m-1$. From here we can see that if $\Delta t$ is constant as in DT-VAR, the model is simply a CT-VAR model where $\Delta t$ is constant. As such, the DT-VAR model is a special case of the CT-VAR model.

Let us look at the link between the CT-VAR and the DT solution parameters. Whereas, the parameter $\boldsymbol{\Phi}$ describes the moment-to-moment association of one variable on the rate of change in another, the parameter $\boldsymbol{\beta}$ is a matrix of lagged parameters, that is, the effect of the variable on a previous measurement occasion to the current measurement occasion, given a particular time interval $\Delta t$. The DT $\boldsymbol{\beta}$ can be obtained from the CT $\boldsymbol{\Phi}$ using

$$\boldsymbol{\beta} = \exp\left(\Delta t \boldsymbol{\Phi}\right). \tag{6}$$

Furthermore, the DT parameter $\boldsymbol{\alpha}$ is related to the CT parameters $\boldsymbol{\Phi}$ and $\boldsymbol{\mu}$

$$\begin{aligned} \boldsymbol{\alpha} &= \left(\boldsymbol{\Phi}^{-1}\right)\left(\exp\left(\Delta t \boldsymbol{\Phi}\right) - \mathbf{I}\right)\left(-\boldsymbol{\Phi}\boldsymbol{\mu}\right) \\ &= \left(\boldsymbol{\Phi}^{-1}\right)\left(\boldsymbol{\beta} - \mathbf{I}\right)\left(-\boldsymbol{\Phi}\boldsymbol{\mu}\right). \end{aligned} \tag{7}$$

If we solve for $\boldsymbol{\mu}$ in Equation 7, we arrive at

$$\boldsymbol{\mu} = \left(\boldsymbol{I} - \boldsymbol{\beta}\right)^{-1}\boldsymbol{\alpha}. \tag{8}$$

Assuming weak stationarity, the mean vector of $\boldsymbol{\eta}$ as a function of the DT parameters $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ is given by $\boldsymbol{\mu}_{\boldsymbol{\eta}} = \left(\mathbf{I} - \boldsymbol{\beta}\right)^{-1}\boldsymbol{\alpha}$. This is equivalent to the CT-VAR parameter $\boldsymbol{\mu}$ as a function of $\boldsymbol{\alpha}$ and $\boldsymbol{\beta}$ given by Equation 8. Lastly, we can convert the CT process noise covariance matrix $\boldsymbol{\Sigma}$ to the DT counter part $\boldsymbol{\Psi}$[3]. Note that $\mathbf{J}$ and $\mathbf{I}$ are identity matrices and $\mathrm{vec}\left(\cdot\right)$ stacks the values from a matrix into a vector by column.

$$\begin{aligned} \mathrm{vec}\left(\boldsymbol{\Psi}\right) &= \mathbf{C}\left[\exp\left(\Delta t \mathbf{C}\right) - \mathbf{J}\right]\mathrm{vec}\left(\boldsymbol{\Sigma}\right) \\ \text{with} \quad \mathbf{C} &= \left(\boldsymbol{\Phi} \otimes \mathbf{I}\right) + \left(\mathbf{I} \otimes \boldsymbol{\Phi}\right) \end{aligned} \tag{9}$$

In summary, we can see the DT-VAR parameters are related to CT-VAR parameters. We can also see from the discrete state space integral solution of CT-VAR that DT-VAR is a special case of CT-VAR where the time interval $\Delta t$ is constant across time and individuals. As $\Delta t$ is part of the calculation of the parameters $\alpha$, $\beta$, and $\Psi$, their values change as a function of the time interval.

## Quantifying Uncertainty in Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

The work of Deboeck and Preacher (2015) provided a starting point for structuring mediation processes in CT-VAR. However, their work is far from complete: complete inferences require the availability of reasonable standard error estimates and ways of quantifying the uncertainty around the direct, indirect, and total effects in mediation in continuous time. In the mediation literature, it is not enough to have point estimates of the indirect effects. It is important to quantify the uncertainty associated with it. Although the idea

[3]The notation presented here is mathematically equivalent to the notation presented in the works of Arnold (1974), Oud and Jansen (2000), and Voelkle and Oud (2012). We prefer this notation because it facilitates straightforward calculations in statistical packages that support standard linear algebra operations.

of the indirect effect as a product of regression coefficients is straightforward, quantifying the uncertainty of the point estimates has some complications and much of the mediation literature has focused on this task. The sampling distribution of the product of normally distributed random variables is not necessarily normal (Aroian, 1947; Craig, 1936), leading to complexities in computing $p$-values for the indirect effect. Because of this, confidence intervals have been used to quantify uncertainty for the point estimates. Using the delta method, confidence intervals that are asymptotically normally distributed have been proposed (Aroian, 1947; Goodman, 1960; Sobel, 1982, 1986, 1987). Alternative approaches that take into account the asymmetry have also been proposed including: (1) The distribution of the product method, which analytically examines the sampling distribution of the product of normally distributed coefficients (MacKinnon et al., 2004, 2007); (2) Nonparametric bootstrapping, involving resampling the original data and fitting the model multiple times to create a sampling distribution of the indirect effect (Bollen & Stine, 1990; Preacher & Hayes, 2008; Shrout & Bolger, 2002); (3) Parametric bootstrapping, involving simulating raw data using the estimated parameters and fitting the model similar to nonparametric bootstrapping (Preacher & Selig, 2012); (4) MC, which simulates regression coefficients to generate a sampling distribution of the indirect effect (MacKinnon et al., 2004; Pesigan & Cheung, 2023; Preacher & Selig, 2012); (5) Profile likelihood-based approach that estimate confidence intervals using the likelihood ratio test (Cheung, 2009a, 2009b; Pawitan, 2013; Pesigan & Cheung, 2020; Venzon & Moolgavkar, 1988). From these approaches, nonparametric bootstrap has become the preferred method (Bollen & Stine, 1990; Preacher & Hayes, 2008; Shrout & Bolger, 2002). However, the multiple model fitting especially for more complex models can be a bottleneck making it a less viable approach (see Pesigan & Cheung, 2023, for the case of mediation models with missing data). MC has been shown to be as good as a nonparametric bootstrap in many situations (Pesigan & Cheung, 2023; Tofighi & Kelley, 2019, 2020). Because the empirical sampling distributions of parameter estimates are generated and not obtained via multiple model fitting, the MC is computationally more efficient than nonparametric bootstrapping.

In this manuscript, drawing from the mediation literature (e.g., MacKinnon et al., 2004; Pesigan & Cheung, 2023; Preacher & Selig, 2012; Sobel, 1986), we provide estimation procedures to complete the inferential process in CT-VAR mediation using the delta and MC methods. Additionally, we provide the `R` package `cTMed` that calculates point estimates, standard errors, and confidence intervals for the direct, indirect, and total effects. The package also provides plotting tools to visualize how the effects change given a range of time intervals.

## Hypothetical Examples

In this section, we use hypothetical examples within the CT-VAR framework to (1) elaborate on what we mean by direct, indirect, and total effects within this modeling framework, (2) develop the delta and MC methods for generating standard errors and confidence intervals for the effects previously defined, and (3) introduce regions of significance using the delta and MC methods to visualize statistically significant results across a range of time intervals.

**Hypothetical Example 1: Understanding Direct, Indirect, and Total Mediation Effects in CT-VAR Models**

In this example, we will illustrate how to compute direct, indirect, and total effects from the estimated drift matrix $\boldsymbol{\Phi}$ in a three-variable CT-VAR model with bidirectional influences. Furthermore, we will examine how these effects change as a function of time interval. In both cross-sectional and autoregressive models, the fundamental concept of mediation analysis involves decomposing the total effect into direct and indirect components. This approach allows us to understand how the mediator variable $M$ influences the relationship between $X$ and $Y$. However, in the context of autoregressive models, several authors (e.g., Aalen et al., 2012, 2016; Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) propose a conceptual shift when thinking about the lagged parameters $\boldsymbol{\beta}$. In this conceptualization, $\boldsymbol{\beta}$ is treated as the matrix of total effects rather than direct effects. This shift has implications for thinking about mediation models in VAR models particularly when it comes to decomposing the total effect into direct and indirect effects.

In the following illustrative example, we use simulated data from descriptive statistics reported by Grundy et al. (2007) to demonstrate how the direct and indirect effects are decomposed from the total effect. We used the `mvrnorm` function from the `MASS R` package (Venables & Ripley, 2002) to generate multivariate normal data using the correlation matrix reported in Table 1 of Grundy et al. (2007). The researchers tested the hypothesis that maternal knowledge mediated the association between mother-reported marital conflict and child-reported behavioral competence. Furthermore, they added a reciprocal effect from behavioral competence to knowledge. The original data set was collected using a discrete time interval of a year over four years. Note that Deboeck and Preacher (2015) also use the descriptive statistics from Grundy et al. (2007) to demonstrate CT mediation. While our approach is similar to that of Deboeck and Preacher (2015), for illustrative purposes, the model we fitted has some differences. First, we fitted a CT-VAR model in in the `dynr R` package (Ou et al., 2019) using the simulated data set where both the measurement and the dynamic structures are estimated. Second, we specified all possible paths from one variable to another to be estimated in the drift matrix $\boldsymbol{\Phi}$. Third, instead of using the empirical sample size of 133, we used different sample sizes including 100, 200, 500, and 1000 to show the effect of sample size on the results[4]. Prior to fitting the model, we inserted missing values in between measurement occasions discrete measurement occasions using .10 increments to aid in the optimization process (Singer, 2012; Voelkle & Oud, 2012).

We label the independent variable marital conflict $X$, the mediator variable maternal knowledge $M$, and dependent variable behavioral competence $Y$. Since, we are incorporating a measurement model, the latent variables $\eta_X$, $\eta_M$, and $\eta_Y$ represent the latent variables in the model with $X$, $M$, and $Y$ as observed indicators. The estimated CT-VAR drift matrix $\boldsymbol{\Phi}$ are presented in Table 1. In the study, time $t$ is scaled in years, and $\Delta t = 1$ a time interval of one year between measurement occasions given by Figure 2a. Given Equation 6 we can calculate the matrix of lagged coefficients for a time interval $\Delta t = 1$ from the drift matrix $\boldsymbol{\Phi}$ which is also presented in Table 1 as follows:

[4]We used the argument `empirical = TRUE` in `mvrnorm` for the function to reproduce the specified `Sigma` for all simulated data. This makes the estimated drift matrix $\boldsymbol{\Phi}$ the same regardless of the sample size used.

**Table 1**

*Point Estimates and 95% Confidence Intervals of the CT-VAR Drift Matrix* $\boldsymbol{\Phi}$ *and the DT-VAR* ($\Delta t = 1$) *Lagged Coefficients Matrix* $\boldsymbol{\beta}$ *for Data Simulated from Grundy et al. (2007)*

| Parameter | Estimate | 2.5% | 97.5% | Estimate | 2.5% | 97.5% | 2.5% | 97.5% |
|---|---|---|---|---|---|---|---|---|
| | $CT-VAR$ | | | DT-VAR | | | | |
| | | | | | Delta | | MC | |
| Conflict to conflict ($X \to X$) | −0.033 | −0.050 | −0.017 | 0.965 | 0.950 | 0.980 | 0.950 | 0.981 |
| Conflict to knowledge ($X \to M$) | −0.152 | −0.195 | −0.108 | −0.121 | −0.152 | −0.090 | −0.151 | −0.089 |
| Conflict to competence ($X \to Y$) | −0.100 | −0.141 | −0.060 | −0.083 | −0.113 | −0.052 | −0.113 | −0.052 |
| Knowledge to conflict ($M \to X$) | 0.038 | 0.011 | 0.065 | 0.028 | 0.008 | 0.047 | 0.009 | 0.047 |
| Knowledge to knowledge ($M \to M$) | −0.585 | −0.711 | −0.460 | 0.562 | 0.493 | 0.630 | 0.497 | 0.634 |
| Knowledge to competence ($M \to Y$) | 0.104 | 0.027 | 0.181 | 0.059 | 0.016 | 0.102 | 0.015 | 0.100 |
| Competence to conflict ($Y \to X$) | −0.013 | −0.037 | 0.011 | −0.007 | −0.024 | 0.010 | −0.024 | 0.010 |
| Competence to knowledge ($Y \to M$) | 0.228 | 0.147 | 0.309 | 0.133 | 0.091 | 0.175 | 0.090 | 0.173 |
| Competence to competence ($Y \to Y$) | −0.511 | −0.616 | −0.406 | 0.607 | 0.547 | 0.667 | 0.550 | 0.671 |

*Note.* Estimates for the CT-VAR are calculated using the `dynr` package. Estimates for the DT-VAR are calculated using the `DeltaBeta` (delta method) and `MCBeta` (MC) functions from the `cTMed` package. These results are based on simulated data with $n = 1,000$.

$$
\begin{aligned}
\boldsymbol{\beta}_{\Delta t(1)} &= \exp\left(\boldsymbol{\Phi}\right) \\
&= \begin{array}{c} \\ \eta_{X_{t+1}} \\ \eta_{M_{t+1}} \\ \eta_{Y_{t+1}} \end{array} \begin{array}{c} \begin{array}{ccc} \eta_{X_t} & \eta_{M_t} & \eta_{X_t} \end{array} \\ \begin{pmatrix} \beta_{1,1} & \beta_{1,2} & \beta_{1,3} \\ \beta_{2,1} & \beta_{2,2} & \beta_{2,3} \\ \beta_{3,1} & \beta_{3,2} & \beta_{3,3} \end{pmatrix} \end{array}.
\end{aligned} \tag{10}
$$

Now we look at the case of a DT-VAR model from the same continuous process except that the time interval used is three years ($\Delta t = 3$) between measurement occasions. From the time-series literature (e.g., Hamilton, 1994) we can derive $\boldsymbol{\beta}_{\Delta t(3)}$ from $\boldsymbol{\beta}_{\Delta t(1)}$ using matrix powers. The key is to raise the matrix with the smaller time interval by the factor in which the time interval increased to the larger time interval. For example, the change from $\Delta t = 1$ to $\Delta t = 3$ is a factor of three, thus $\boldsymbol{\beta}^3_{\Delta t(1)} = \boldsymbol{\beta}_{\Delta t(3)}$. Therefore, we can say that $\boldsymbol{\beta}_{\Delta t(3)} = \exp\left(3\boldsymbol{\Phi}\right) = \boldsymbol{\beta}^3_{\Delta t(1)}$. Let us explore the relationship of the elements of $\boldsymbol{\beta}_{\Delta t(3)}$ as a function of the elements of $\boldsymbol{\beta}_{\Delta t(1)}$. In particular, we will focus on the third row and first column in $\boldsymbol{\beta}_{\Delta t(3)}$ pertaining to the association of marital conflict ($\eta_X$) at time $t$ and behavioral competence ($\eta_Y$) at time $t+3$. The purple line in Panel 2a in Figure 2 represents the total effect of marital conflict ($\eta_X$) at time $t$ on behavioral competence ($\eta_Y$) at time $t+3$. This total effect is decomposed as a function of the lagged parameters in $\boldsymbol{\beta}_{\Delta t(1)}$. By raising Equation 10 to the third power and subsetting the third row and the first column of the resulting matrix we get

$$
\begin{aligned}
\text{Total} &= \boldsymbol{\beta}_{\Delta t(3)}\left[3,1\right] \\
&= \beta_{1,1}\beta_{1,1}\beta_{3,1} + \beta_{2,1}\beta_{1,2}\beta_{3,1} + \beta_{3,1}\beta_{1,3}\beta_{3,1} \\
&+ \beta_{1,1}\beta_{2,1}\beta_{3,2} + \beta_{2,1}\beta_{2,2}\beta_{3,2} + \beta_{3,1}\beta_{2,3}\beta_{3,2} \\
&+ \beta_{1,1}\beta_{3,1}\beta_{3,3} + \beta_{2,1}\beta_{3,2}\beta_{3,3} + \beta_{3,1}\beta_{3,3}\beta_{3,3},
\end{aligned} \tag{11}
$$

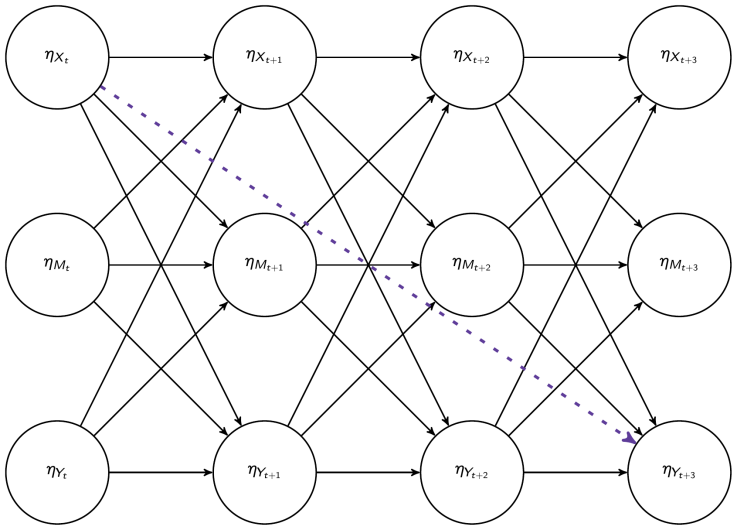
where each term represents distinct paths from marital conflict ($\eta_X$) at time $t$ to behavioral competence ($\eta_Y$) at time $t+3$ graphically represented by Panels 2b to 2j in Figure 2. Note that red paths are paths where maternal knowledge ($\eta_M$) is traversed. We define these red paths as indirect effects of marital conflict ($\eta_X$) at time $t$ on behavioral competence ($\eta_Y$) at time $t+3$ through maternal knowledge ($\eta_M$). The sum of these red paths constitutes the overall indirect effect. The blue paths are the direct effects and their sum constitutes the overall direct effect of marital conflict ($\eta_X$) at time $t$ on behavioral competence ($\eta_Y$) at time $t+3$. The sum of the direct and indirect effects represents the total effect of marital conflict ($\eta_X$) at time $t$ on behavioral competence ($\eta_Y$) at time $t+3$.

Now we divide any arbitrary time interval $\Delta t$ into infinitely many smaller parts and insert these infinitely many latent variables between values at time $t$ and time $t+\Delta t$. The idea in the CT-VAR approach is that the lagged parameters from the DT integral solution represents the sum of all the effects from the infinitely many smaller parts in between time $t$ and time $t+\Delta t$. The decomposition we discussed earlier still apply. How do we then obtain the direct, indirect, and total effects? The total effect is straight forward as it is simply the result of exponentiating the product of the drift matrix and the time interval.

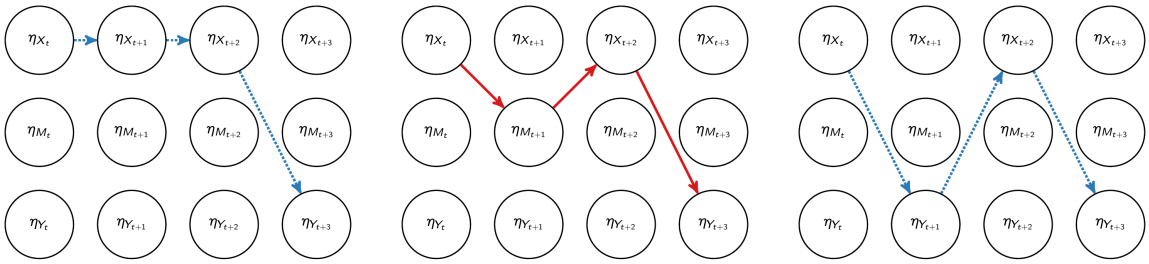
**Figure 2**

*Decomposing the Direct and Indirect Effects from the Total Effect*

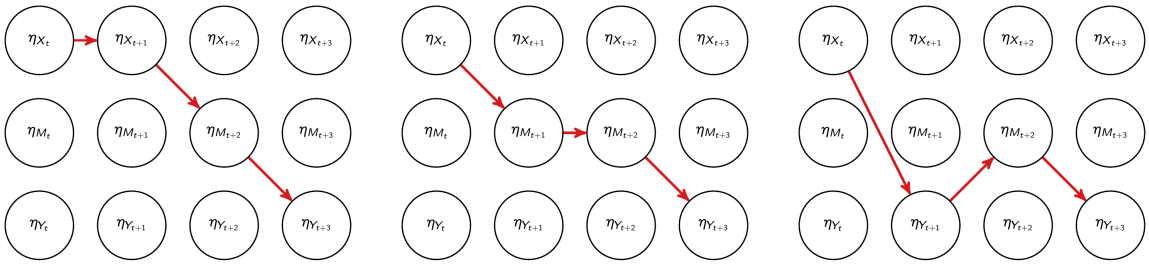
(a) *DT-VAR with time interval of one. The purple line represents the total effect of $X$ on $Y$ at a time interval of three.*



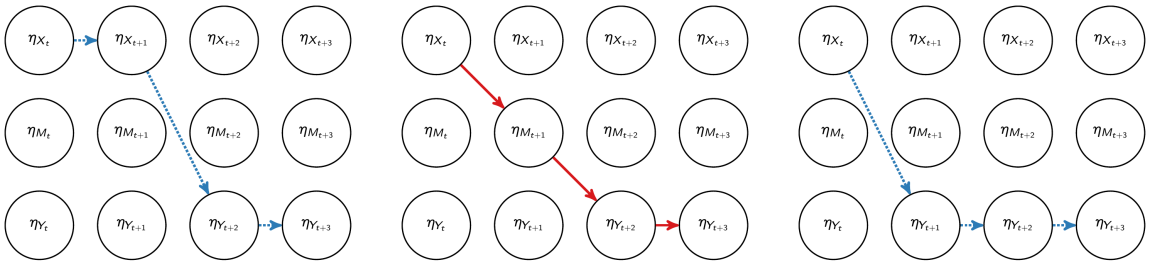

(b) $\beta_{1,1}\beta_{1,1}\beta_{3,1}$ (c) $\beta_{2,1}\beta_{1,2}\beta_{3,1}$ (d) $\beta_{3,1}\beta_{1,3}\beta_{3,1}$



(e) $\beta_{1,1}\beta_{2,1}\beta_{3,2}$ (f) $\beta_{2,1}\beta_{2,2}\beta_{3,2}$ (g) $\beta_{3,1}\beta_{2,3}\beta_{3,2}$



(h) $\beta_{1,1}\beta_{3,1}\beta_{3,3}$ (i) $\beta_{2,1}\beta_{3,2}\beta_{3,3}$ (j) $\beta_{3,1}\beta_{3,3}\beta_{3,3}$



*Note:* Panel (2a) represents the total effect of $\eta_X$ on $\eta_Y$ at a time interval of three is a function of the lagged parameters for a time interval of one. Panels (2b) to (2j) represent the decomposition of the total effect (purple) into direct (blue) and indirect (red) effects.

Using the estimated drift matrix presented in Table 1 and a time interval $\Delta t = 3$, the total effect matrix is given by

$$\begin{aligned}\text{Total} &= \exp\left(\Delta t \boldsymbol{\Phi}\right) \\ &= \exp\left[3\begin{pmatrix} -0.033 & 0.038 & -0.013 \\ -0.152 & -0.585 & 0.228 \\ -0.100 & 0.104 & -0.511 \end{pmatrix}\right] \\ &= \begin{pmatrix} 0.892 & 0.049 & -0.006 \\ -0.240 & 0.183 & 0.139 \\ -0.172 & 0.056 & 0.239 \end{pmatrix}.\end{aligned} \tag{12}$$

Following Ryan and Hamaker (2021), the matrix of direct effects is obtained by removing the effects in the drift matrix $\boldsymbol{\Phi}$ that originates from and/or goes to the mediator (or mediators) before exponentiating. We do this by pre- and post-multiplying the drift matrix $\boldsymbol{\Phi}$ with a diagonal matrix $\mathbf{D_m}$ where the $i^{\text{th}}$ and $j^{\text{th}}$ index pertaining to the mediators are set to zero. In our current example where maternal knowledge ($\eta_M$) is treated as the mediator variable, the direct effect matrix is given by

$$\begin{aligned}\text{Direct} &= \exp\left(\Delta t \boldsymbol{D_m \Phi D_m}\right) \\ &= \exp\left[3\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} -0.033 & 0.038 & -0.013 \\ -0.152 & -0.585 & 0.228 \\ -0.100 & 0.104 & -0.511 \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}\right] \\ &= \begin{pmatrix} 0.908 & 0.000 & -0.019 \\ 0.000 & 1.000 & 0.000 \\ -0.145 & 0.000 & 0.218 \end{pmatrix}.\end{aligned} \tag{13}$$

The indirect effect can now be obtained by subtracting the direct effect matrix from the total effect matrix.

$$\begin{aligned}\text{Indirect} &= \text{Total} - \text{Direct} \\ &= \exp\left(\Delta t \boldsymbol{\Phi}\right) - \exp\left(\Delta t \boldsymbol{D_m \Phi D_m}\right) \\ &= \begin{pmatrix} -0.016 & 0.049 & 0.013 \\ -0.240 & -0.817 & 0.139 \\ -0.027 & 0.056 & 0.021 \end{pmatrix}\end{aligned} \tag{14}$$

We subset the $i^{\text{th}}$ row and $j^{\text{th}}$ column of the resulting matrices for the direct, indirect, and total effects of the mediator variable/s on the path from the $j^{\text{th}}$ column variable (1 for marital conflict in our case) to the $i^{\text{th}}$ row variable (3 for behavioral competence in our case). The direct, indirect, and total effects of marital conflict ($\eta_X$) on behavioral competence ($\eta_Y$) through maternal knowledge ($\eta_M$), for a time interval of 3, are $-0.145$, $-0.027$, and $-0.172$, respectively[5].

[5]The `Med` function in the `cTMed R` package calculates the direct, indirect, and total effects for a specific time interval or a range of time intervals. A plot method for the `Med` function can be used to visualize the effects as a function of time interval.

The sum of all the effects that pass from marital conflict to behavioral competence, which we call the total effect, is $-0.172$. This is the accumulated effect up until the third year (given $\Delta t = 3$ and time $t$ scaled in terms of years). We can decompose this total effect in terms of the effect that pass from marital conflict to behavioral competence without passing through maternal knowledge which is the direct effect, that is, $-0.145$ as well as the effects that pass from marital conflict to behavioral competence through maternal knowledge which is the indirect effect, that is, $-0.027$. From this decomposition, we can see based on the absolute values of the direct and indirect effect, that much of the effect of maternal conflict on behavioral competence is direct but there are also effects that pass thorough maternal knowledge.

## Illustrative Example 2: Quantifying Uncertainty in Direct, Indirect, and Total Mediation Effects in CT-VAR Models

In the previous section, we showed how to calculate point estimates for the direct, indirect, and total effect for a given time interval from the CT-VAR drift matrix $\boldsymbol{\Phi}$. In particular, we know the direct, indirect, and total effects of marital conflict ($\eta_X$) on behavioral competence ($\eta_Y$) through maternal knowledge ($\eta_M$), for a time interval of 3, are $-0.145$, $-0.027$, and $-0.172$, respectively. However, we still do not have a way to assess these effects from an inferential point of view. In this section, we extend the previous section by providing methods for quantifying the uncertainty in the direct, indirect, and total effects we have defined. In the following sections, we introduce the delta and the MC methods. We chose these approaches because they are relatively easy to compute even for complex models. We also wish to contrast the performance of a method where symmetry is imposed versus and a method where asymmetry is accounted for. We discuss these methods in general and then apply them to the context of generating standard errors and confidence intervals as a way to quantify the uncertainty for the direct, indirect, and total effects.

### *The Delta Method*

The delta method is a powerful statistical technique used to derive the asymptotic distribution of a random variable. It becomes particularly handy when dealing with functions of other random variables that are asymptotically Gaussian, that is, their distributions converge to a Gaussian distribution as sample size increases. As previously mentioned, it has been applied in psychological research to generate standard errors and confidence intervals for the indirect effect (Aroian, 1947; Goodman, 1960; Sobel, 1986, 1987). It has also been applied to other functions of parameters such as the standardized regression coefficients (Jones & Waller, 2015; Pesigan et al., 2023; K.-H. Yuan & Chan, 2011). Currently, popular SEM software packages such as `Mplus` (Muthén & Muthén, 2017) and `lavaan` (Rosseel, 2012) use the delta method as the standard approach to generating standard errors and confidence intervals for defined functions of SEM parameters. In `R`, the `Delta` and `DeltaGeneric` functions in the `betaDelta` (Pesigan et al., 2023) package can generate standard errors and confidence intervals for functions of parameters of fitted models as long as the estimates and the sampling variance-covariance matrix are provided.

We define the delta method following the multivariate central limit theorem. Let $\boldsymbol{\theta}$ be $\text{vec}\,(\boldsymbol{\Phi})$, that is, the elements of the drift matrix $\boldsymbol{\Phi}$ in vector form sorted column-wise.

Let $\hat{\boldsymbol{\theta}}$ be vec $\left(\hat{\boldsymbol{\Phi}}\right)$, that is the estimates of the elements of the drift matrix. The uncertainty around any functions of the estimates for $\hat{\boldsymbol{\theta}}$ is assumed to be characterized as:

$$\sqrt{n}\left(\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)-\mathbf{g}\left(\boldsymbol{\theta}\right)\right)\xrightarrow{\mathrm{D}}\mathcal{N}\left(0,\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'\right) \tag{15}$$

where $\mathbf{g}$ is a vector of functions, $\mathbf{J}$ is the matrix of first-order derivatives of the functions in $\mathbf{g}$ with respect to the elements of $\boldsymbol{\theta}$ and $\boldsymbol{\Gamma}$ is the asymptotic variance-covariance matrix of $\hat{\boldsymbol{\theta}}$. From the former, we can derive the distribution of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ for a sample size $n$ as follows

$$\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)\approx\mathcal{N}\left(\mathbf{g}\left(\boldsymbol{\theta}\right),n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'\right). \tag{16}$$

The uncertainty associated with the estimator $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ is, therefore, given by $n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'$ . When $\boldsymbol{\Gamma}$ is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of $\hat{\boldsymbol{\theta}}$, that is, $\hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)$ for $n^{-1}\boldsymbol{\Gamma}$. Therefore, the sampling variance-covariance matrix of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ is given by

$$\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)\approx\mathcal{N}\left(\mathbf{g}\left(\boldsymbol{\theta}\right),\mathbf{J}\hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)\mathbf{J}'\right). \tag{17}$$

From this general approach, we can generate the sampling variance-covariance matrix of any function of the drift matrix $\boldsymbol{\Phi}$. The standard errors, that is, the square root of the diagonal elements of the sampling variance-covariance matrix are used to generate symmetric confidence intervals around the point estimates of any function of the drift matrix $\boldsymbol{\Phi}$. In the current application, let the function $\mathbf{g}$ be a vector of the direct, indirect, and total effects of a specific independent variable on a dependent variable through one or more mediator variables. The `DeltaMed` function in the `cTMed` package applies the delta method to generate standard errors, test statistics, $p$-values, and confidence intervals for these effects for a specific time interval or a range of time intervals.

### *The Monte Carlo Method*

MC was introduced in the mediation literature by MacKinnon et al. (2004) and formalized by Preacher and Selig (2012) as a resampling technique for generating confidence intervals for the indirect effect. It has been shown to be as good as nonparametric bootstrap (Pesigan & Cheung, 2023; Tofighi & Kelley, 2019, 2020; Tofighi & MacKinnon, 2015). As a resampling technique, it is similar to the bootstrap such that the sampling distribution of the function of parameters does not need to be normal. Unlike the bootstrap, however, it is more computationally efficient as it does not involve multiple model fitting. The `R` package `semmcci` provides a general approach to applying MC for generating confidence intervals for a function of any SEM parameters. The `MCFunc` and `MCGeneric` functions in the `semmcci` (Pesigan & Cheung, 2023) package can generate standard errors and confidence intervals for functions of parameters of fitted models as long as the estimates and the sampling variance-covariance matrix are provided.

Based on the asymptotic properties of maximum likelihood estimators, we can assume that they are normally distributed around the population parameters.

$$\hat{\boldsymbol{\theta}}\sim\mathcal{N}\left(\boldsymbol{\theta},\mathbb{V}\left(\hat{\boldsymbol{\theta}}\right)\right) \tag{18}$$

Using this distributional assumption, a sampling distribution of $\hat{\boldsymbol{\theta}}$ which we refer to as $\hat{\boldsymbol{\theta}}^*$ can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\boldsymbol{\theta}}^* \sim \mathcal{N}\left(\hat{\boldsymbol{\theta}}, \hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)\right). \tag{19}$$

A sampling distribution of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ , which we refer to as $\mathbf{g}\left(\hat{\boldsymbol{\theta}}^*\right)$ , can be generated by using the simulated estimates to calculate $\mathbf{g}$. The standard deviations of the simulated estimates are the standard errors. In resampling approaches, it is customary to use the percentiles of the generated distribution as confidence intervals instead of the standard errors. This approach allows the confidence limits to be asymmetric around the point estimates unlike the delta method which is always symmetric. Percentiles corresponding to $100\,(1-\alpha)\,\%$ are the confidence intervals.

From this general approach, we can generate the empirical sampling distribution of any function of the drift matrix $\boldsymbol{\Phi}$. There are constraints, however, on the randomly generated values of $\boldsymbol{\Phi}$. First, the diagonal values should be from zero to negative infinity. Second, the real part of all eigenvalues is less than zero. These two can be implemented using a replacement sampling approach, that is, discarding randomly generated values where the two constraints do not hold. Again, let the function $\mathbf{g}$ be a vector of the direct, indirect, and total effects of a specific independent variable on a dependent variable through one or more mediator variables. The `MCMed` function in the `cTMed` package applies MC to generate the standard errors and confidence intervals for these effects for a specific time interval or a range of time intervals.

### *Confidence Intervals for the Direct, Indirect and Total Effects*

Using the delta and MC methods as implemented by the `cTMed` package, the standard errors and confidence intervals for the direct, indirect, and total effects of marital conflict ($\eta_X$) on behavioral competence ($\eta_Y$) through maternal knowledge ($\eta_M$), for time intervals $\Delta t$ of 1, 2, and 3 are presented in Table 2. We chose these time intervals as they are the time intervals available in the data set. Based on the results we highlight a few observations. The direct, indirect, and total effects change across time intervals. In fact, the absolute value of the direct, indirect, and total effects increase as time intervals increase implying stronger effects at larger time gaps between measurements.

Write about the confidence intervals as a function of sample sizes Write about the difference between delta and MC in terms of symmetry

Looking at the confidence intervals, however, we can see that the direct and total effects are not statistically different from zero while the indirect effect is significant. We can see this as the confidence intervals for the indirect effect for all three time intervals did not contain zero. From a substantive point of view, this shows that the association between marital conflict ($\eta_X$) and behavioral competence ($\eta_Y$) is mediated by maternal knowledge ($\eta_M$) given time intervals of one, two, and three years.

## Illustrative Example 3: Using Regions of Significance in Direct, Indirect, and Total Effects to Inform Designs of Longitudinal Mediation Studies

The Johnson-Neyman technique (Bauer & Curran, 2005; Johnson & Fay, 1950; Johnson & Neyman, 1936; Preacher et al., 2006) is a statistical method utilized for analyzing

**Table 2**

*Point Estimates, Standard Errors, and 95% Confidence Intervals of the Direct, Indirect, and Total Effects for Data Simulated from Grundy et al. (2007)*

| $\Delta t$ | Effect | Estimate | SE | 2.5% | 97.5% | Sym* | SE | 2.5% | 97.5% | Sym* |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | Delta | | | | MC | | | |
| | | | $n = 100$ | | | | | | | |
| 1 | Direct | −0.077 | 0.050 | −0.176 | 0.022 | 0.000 | 0.050 | −0.177 | 0.021 | 0.002 |
| 1 | Indirect | −0.006 | 0.007 | −0.020 | 0.008 | 0.000 | 0.007 | −0.022 | 0.008 | 0.002 |
| 1 | Total | −0.083 | 0.049 | −0.180 | 0.014 | 0.000 | 0.050 | −0.179 | 0.014 | 0.001 |
| 2 | Direct | −0.121 | 0.080 | −0.278 | 0.037 | 0.000 | 0.081 | −0.287 | 0.033 | 0.012 |
| 2 | Indirect | −0.017 | 0.020 | −0.056 | 0.022 | 0.000 | 0.021 | −0.059 | 0.026 | 0.000 |
| 2 | Total | −0.137 | 0.077 | −0.289 | 0.014 | 0.000 | 0.078 | −0.289 | 0.017 | 0.003 |
| 3 | Direct | −0.144 | 0.098 | −0.336 | 0.047 | 0.000 | 0.101 | −0.360 | 0.039 | 0.032 |
| 3 | Indirect | −0.027 | 0.032 | −0.091 | 0.036 | 0.000 | 0.035 | −0.095 | 0.047 | 0.007 |
| 3 | Total | −0.172 | 0.093 | −0.354 | 0.011 | 0.000 | 0.096 | −0.363 | 0.017 | 0.003 |
| | | | $n = 200$ | | | | | | | |
| 1 | Direct | −0.077 | 0.036 | −0.147 | −0.007 | 0.000 | 0.036 | −0.148 | −0.007 | 0.000 |
| 1 | Indirect | −0.006 | 0.005 | −0.015 | 0.004 | 0.000 | 0.005 | −0.016 | 0.004 | 0.001 |
| 1 | Total | −0.083 | 0.035 | −0.151 | −0.014 | 0.000 | 0.035 | −0.151 | −0.014 | 0.001 |
| 2 | Direct | −0.121 | 0.057 | −0.232 | −0.010 | 0.000 | 0.057 | −0.237 | −0.011 | 0.006 |
| 2 | Indirect | −0.017 | 0.014 | −0.044 | 0.011 | 0.000 | 0.014 | −0.046 | 0.012 | 0.000 |
| 2 | Total | −0.137 | 0.055 | −0.244 | −0.030 | 0.000 | 0.055 | −0.245 | −0.029 | 0.001 |
| 3 | Direct | −0.145 | 0.069 | −0.280 | −0.009 | 0.000 | 0.071 | −0.292 | −0.013 | 0.016 |
| 3 | Indirect | −0.027 | 0.023 | −0.072 | 0.018 | 0.000 | 0.024 | −0.073 | 0.022 | 0.003 |
| 3 | Total | −0.172 | 0.066 | −0.301 | −0.042 | 0.000 | 0.067 | −0.304 | −0.040 | 0.001 |
| | | | $n = 500$ | | | | | | | |
| 1 | Direct | −0.077 | 0.023 | −0.121 | −0.033 | 0.000 | 0.023 | −0.122 | −0.033 | 0.000 |
| 1 | Indirect | −0.006 | 0.003 | −0.012 | 0.000 | 0.000 | 0.003 | −0.012 | 0.000 | 0.001 |
| 1 | Total | −0.083 | 0.022 | −0.126 | −0.040 | 0.000 | 0.022 | −0.126 | −0.039 | 0.001 |
| 2 | Direct | −0.121 | 0.036 | −0.191 | −0.051 | 0.000 | 0.036 | −0.193 | −0.051 | 0.002 |
| 2 | Indirect | −0.016 | 0.009 | −0.034 | 0.001 | 0.000 | 0.009 | −0.034 | 0.001 | 0.000 |
| 2 | Total | −0.137 | 0.035 | −0.205 | −0.070 | 0.000 | 0.035 | −0.205 | −0.069 | 0.001 |
| 3 | Direct | −0.145 | 0.044 | −0.231 | −0.059 | 0.000 | 0.044 | −0.235 | −0.061 | 0.007 |
| 3 | Indirect | −0.027 | 0.014 | −0.055 | 0.001 | 0.000 | 0.015 | −0.055 | 0.003 | 0.001 |
| 3 | Total | −0.172 | 0.042 | −0.254 | −0.090 | 0.000 | 0.042 | −0.254 | −0.090 | 0.001 |
| | | | $n = 1,000$ | | | | | | | |
| 1 | Direct | −0.077 | 0.016 | −0.108 | −0.046 | 0.000 | 0.016 | −0.108 | −0.046 | 0.000 |
| 1 | Indirect | −0.006 | 0.002 | −0.010 | −0.001 | 0.000 | 0.002 | −0.010 | −0.001 | 0.000 |
| 1 | Total | −0.083 | 0.016 | −0.113 | −0.052 | 0.000 | 0.016 | −0.113 | −0.052 | 0.001 |
| 2 | Direct | −0.121 | 0.025 | −0.171 | −0.071 | 0.000 | 0.026 | −0.172 | −0.072 | 0.001 |
| 2 | Indirect | −0.016 | 0.006 | −0.029 | −0.004 | 0.000 | 0.006 | −0.029 | −0.004 | 0.000 |
| 2 | Total | −0.137 | 0.024 | −0.185 | −0.089 | 0.000 | 0.025 | −0.185 | −0.089 | 0.001 |
| 3 | Direct | −0.145 | 0.031 | −0.206 | −0.084 | 0.000 | 0.031 | −0.208 | −0.085 | 0.003 |
| 3 | Indirect | −0.027 | 0.010 | −0.047 | −0.007 | 0.000 | 0.010 | −0.047 | −0.006 | 0.001 |
| 3 | Total | −0.172 | 0.030 | −0.230 | −0.114 | 0.000 | 0.030 | −0.229 | −0.114 | 0.000 |

*Note.* Delta and MC standard errors and confidence intervals are calculated using the `DeltaMed` and `MCMed` functions from the `cTMed` package, respectively.

*The symmetry of the confidence intervals is assessed using the absolute value of 2.5% − estimate + 97.5% − estimate. Values greater than zero indicate asymmetry.

interactions in regression models. It addresses the limitations of the simple slopes method by offering a more systematic approach to determining the "regions of significance" for interaction effects. Rather than selecting arbitrary conditional values of the moderator variable, this technique computes specific values of the moderator that result in a significant difference in the relationship between the predictor and the outcome variable. By computing the conditional values corresponding to a particular critical ratio (typically to achieve a $p$-value of .05), the Johnson-Neyman technique delineates the regions of significance on the moderator variable. These regions signify the range of moderator values within which the relationship between the predictor and the outcome variable is statistically significant. We aim to extend this concept from the regression framework to CT mediation. Specifically, we seek to identify regions of significance for the direct, indirect, and total effects–that is, the time interval values where these effects are statistically significant.

The parameters in the DT solution of CT-VAR are defined for any time interval. As such, we can generate a continuous vector of time intervals $\Delta t$ and solve for the direct, indirect, and total effects using the CT-VAR drift matrix $\mathbf{\Phi}$. Using the delta and MC methods, confidence intervals for the vector of time intervals can be generated and visualized. A consequence of this, is the ability to visualize regions of significance for the direct, indirect, and total effects given a range of time interval values analogous to the Johnson-Neyman technique. Whereas, previous research has focused on point estimates and finding the maximal effects (e.g., Deboeck & Preacher, 2015; Hecht & Zitzmann, 2021; Wang & Zhang, 2020), the methods we have developed in previous sections adds the ability to visualize regions where the direct, indirect, and total effects are significantly different from zero.

In Figures 3 and 4 we demonstrate the effect of sample size on the regions of significance. For the sample size of 100, none of the time intervals investigated resulted in statistical significance. Notice also that the confidence intervals are wider. For the sample size of 200, the confidence intervals became narrower and the total effects and a large portion of the direct effect became significant. Notice that a portion of the direct effect was not significant for the delta method. For the sample size of 500, both the direct and total effects are significant. The confidence intervals for the indirect effect has narrowed some more but the upper bound still intersected with zero. For the sample size of $1,000$, the confidence intervals are narrow and all effect are significant across the time intervals examined.

Sample size as a factor for regions of significance

Method as a factor for regions of significance

Substantive interpretation

When $n = 100$ total effect - the lower limit of the confidence intervals above 0 but very close to zero

point estimate of the indirect effect is quite small and close to zero

When $n = 200$

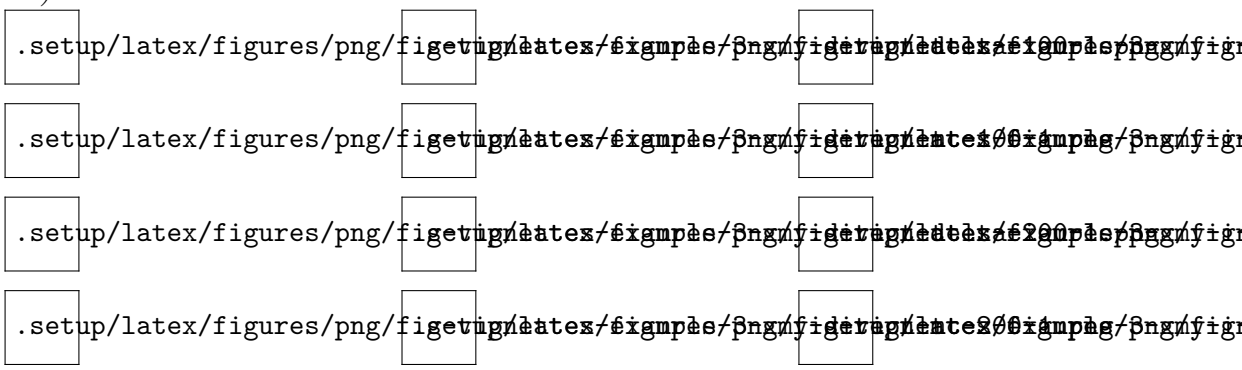
$n = 500$ still not enough to get statistically significant indirect effects

Delta and MC converge when sample size increase

The regions of significance is a great tool for determining the optimality of the time intervals in relation to statistically significant findings. It allows us to examine in a visual way the interplay between factors that influence statistical significance such as sample, effect sizes, and even the method used (e.g., delta vs. MC) to create confidence intervals. Insights gained using the regions of significance can help researchers design studies around the time

**Figure 3**

*Regions of Significance for the Direct, Indirect, and Total Effects (Conflict → Knowledge → Competence)*




*Note:* The shaded areas represent regions of significance, that is, instances where the 95% confidence intervals did not contain zero for a given time interval.

intervals where results can be potentially significant. This approach can become a more principled way to determining when to measure the variables of interest.

Add some interpretations of the regions of significance. Report the ranges of significant $\Delta t$ values. How these regions vary with sample sizes.

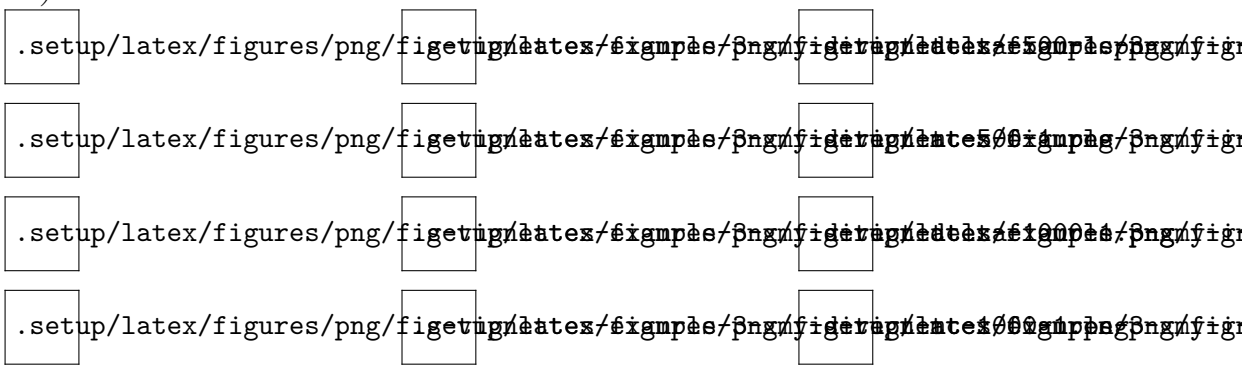
## Monte Carlo Simulation Study

In our simulation study, we investigate the finite sample performance of the delta and MC methods for confidence interval estimation.

Motivate your selection of design factors more. What are they important to consider? Can you cite some previous references to show that they matter?

**Figure 4**

*Regions of Significance for the Direct, Indirect, and Total Effects (Conflict → Knowledge → Competence)*




*Note:* The shaded areas represent regions of significance, that is, instances where the 95% confidence intervals did not contain zero for a given time interval.

MC involves generating $20,000$ drift matrices $\boldsymbol{\Phi}$ for each application. We compare these approaches across different sample sizes (ranging from 50 to 500), various time intervals ($\Delta t$ values from 1 to 30), and different magnitudes of direct, indirect, and total effects (see Table 3). We describe the simulation study setup, including parameter choices, data generation, model fitting, and confidence interval generation. Additionally, we evaluate the performance of both approaches and summarize the simulation results.

## Parameters

Deboeck and Preacher (2015) generated data from a DT-VAR model using a fixed time interval of one unit with the following parameters,

$$\boldsymbol{\beta}_{\Delta t(1)} = \begin{pmatrix} 0.70 & 0.00 & 0.00 \\ 0.50 & 0.60 & 0.00 \\ -0.10 & 0.40 & 0.50 \end{pmatrix},$$

$$\boldsymbol{\alpha}_{\Delta t(1)} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

$$\boldsymbol{\Psi}_{\Delta t(1)} = \begin{pmatrix} 0.10 & 0.00 & 0.00 \\ 0.00 & 0.10 & 0.00 \\ 0.00 & 0.00 & 0.10 \end{pmatrix}$$

and initial values of $\eta_X$, $\eta_M$, and $\eta_Y$ sampled from a standard multivariate normal distribution. Using the DT-VAR parameters above, we derived the CT-VAR parameters following the approach proposed by Chow et al. (2023) arriving at the following

$$\boldsymbol{\Phi} = \begin{pmatrix} -0.36 & 0.00 & 0.00 \\ 0.77 & -0.51 & 0.00 \\ -0.45 & 0.73 & -0.69 \end{pmatrix},$$

$$\boldsymbol{\mu} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

$$\boldsymbol{\Sigma} = \begin{pmatrix} 0.24 & 0.02 & -0.05 \\ 0.02 & 0.07 & 0.02 \\ -0.05 & 0.02 & 0.08 \end{pmatrix}.$$

We used the same mean vector and covariance matrix to generate the initial values of $\eta_X$, $\eta_M$, and $\eta_Y$. In addition to Deboeck and Preacher (2015), we included a measurement model for the observed variables with the following parameters

**Table 3**

*Simulation Parameters*

| $\Delta t$ | Direct Effect | Indirect Effect | Total Effect |
|---|---|---|---|
| 1 | −0.2675 | 0.1674 | −0.1000 |
| 2 | −0.3209 | 0.4008 | 0.0799 |
| 3 | −0.2914 | 0.5423 | 0.2508 |
| 4 | −0.2374 | 0.5823 | 0.3449 |
| 5 | −0.1828 | 0.5521 | 0.3693 |
| 6 | −0.1363 | 0.4846 | 0.3482 |
| 7 | −0.0996 | 0.4037 | 0.3041 |
| 8 | −0.0718 | 0.3242 | 0.2524 |
| 9 | −0.0513 | 0.2533 | 0.2020 |
| 10 | −0.0364 | 0.1938 | 0.1574 |
| 11 | −0.0257 | 0.1459 | 0.1202 |
| 12 | −0.0181 | 0.1085 | 0.0903 |
| 13 | −0.0128 | 0.0798 | 0.0670 |
| 14 | −0.0090 | 0.0582 | 0.0493 |
| 15 | −0.0063 | 0.0422 | 0.0359 |
| 16 | −0.0044 | 0.0304 | 0.0260 |
| 17 | −0.0031 | 0.0218 | 0.0187 |
| 18 | −0.0022 | 0.0156 | 0.0134 |
| 19 | −0.0015 | 0.0111 | 0.0096 |
| 20 | −0.0011 | 0.0079 | 0.0068 |
| 21 | −0.0007 | 0.0056 | 0.0048 |
| 22 | −0.0005 | 0.0040 | 0.0034 |
| 23 | −0.0004 | 0.0028 | 0.0024 |
| 24 | −0.0003 | 0.0020 | 0.0017 |
| 25 | −0.0002 | 0.0014 | 0.0012 |
| 26 | −0.0001 | 0.0010 | 0.0009 |
| 27 | −0.0001 | 0.0007 | 0.0006 |
| 28 | −0.0001 | 0.0005 | 0.0004 |
| 29 | 0.0000 | 0.0003 | 0.0003 |
| 30 | 0.0000 | 0.0002 | 0.0002 |

*Note.* Sample sizes $n$ used include 50, 100, 150, 200, 250, 300, 350, 400, 450, and 500.

$$\boldsymbol{\nu} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

$$\boldsymbol{\Lambda} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$\boldsymbol{\Theta} = \begin{pmatrix} 0.20 & 0.00 & 0.00 \\ 0.00 & 0.20 & 0.00 \\ 0.00 & 0.00 & 0.20 \end{pmatrix}.$$

### Data Generation

We simulated values for three variables—$X$, $M$, and $Y$ for individual $i$ at time $t$ and measurement occasion $m$. The vector of individuals is given by $i = \{1, \cdots, n\}$, where the sample size $n$ was varied using $50, 100, 150, 200, 250, 300, 350, 400, 450$, and 500. The vector of measurement occasions is given by $m = \{0, \cdots, 179\}$, representing daily measurements over 180 days. The vector of time is given by $t = m \times \Delta t$ with $\Delta t = \frac{1}{30}$, representing a day in a month. One hundred observations were randomly selected for each individual. Unselected observations were flagged as missing values, creating irregularly spaced data[6]. Values for the latent variables $\eta_X$, $\eta_M$, and $\eta_Y$ were generated using the DT integral solution of CT-VAR given by Equation 5 with a time interval $\Delta t$ of $\frac{1}{30}$. Observed variables were generated using simulated latent variables and the measurement model given by Equation 1. In both cases, CT-VAR parameters from the previous section were used. The wrapper function `manCTMed::Data` was used to generate data which uses the `SimSSMOUFixed` functiom from the `simStateSpace`[7] `R` package.

### Model Fitting and Confidence Intervals

Parameters of the CT-VAR model were estimated using wrapper function `manCTMed::FitDynr` which uses the `dynr`[8] `R` package. The $\boldsymbol{\Lambda}$ was fixed to a diagnonal matrix. The $\boldsymbol{\mu}$ vector was fixed to a null vector. The off-diagonal elements of the $\boldsymbol{\Theta}$ matrix were fixed to zero and the diagonal elements were estimated. The rest of the parameters were estimated.

The estimated $\boldsymbol{\Phi}$ matrix and its corresponding sampling variance-covariance matrix were extracted from the fitted model. These were used to generate delta and MC method confidence intervals for time intervals of 1 to 30. We looked at two mediation models. The first involves the path from $\eta_X \rightarrow \eta_M \rightarrow \eta_Y$ and the second one involves the path from $\eta_Y \rightarrow \eta_M \rightarrow \eta_X$. For the first model, although the absolute values of all the effects will be close to zero at higher time intervals, we expect the absolute value of the total, direct and indirect effects to be non-zero for values of $\Delta t$ from 1 to 30. For the second model,

[6]Note that the initial value is always included in the sample such that all individuals have the same starting point in time.

[7]See https://CRAN.R-project.org/package=simStateSpace. The specific version used is 1.2.1

[8]The development version available in https://github.com/mhunter1/dynr particularly commit `40277ff6efcfecf04027afa975f7287c6aa81782` was used.

we expect the total, direct and indirect effects to be zero on all values of $\Delta t$. The wrapper function `manCTMed::CI` was used to generate confidence intervals which uses the `DeltaMed` and `MCMed` functions from the `cTMed R` package.

## Results

The Type I error rate represents the proportion of replications where the confidence interval does not contain zero when the population parameter is actually zero. In other words, it is the probability of falsely rejecting the null hypothesis when it is true. We assessed Type I error rates for the mediation model $\eta_Y \rightarrow \eta_M \rightarrow \eta_X$, where the direct, indirect, and total effects are all zero, regardless of the time interval value. Statistical power is the probability of correctly detecting an effect when it actually exists. For the mediation model $\eta_X \rightarrow \eta_M \rightarrow \eta_Y$, where the direct, indirect, and total effects are not zero but decrease as the time interval increases, we evaluated statistical power. Statistical power greater than or equal to 0.80 is considered robust (Cohen, 1988). Coverage probabilities indicate the proportion of replications where the confidence intervals contain the population effect. We considered coverage probabilities within 0.925 and 0.975 as robust for a 95% confidence interval (Bradley, 1978). Detailed results for coverage probabilities, statistical power, and Type I error rates are available in the Supplementary Materials.

### *Coverage Probabilities*

For the delta method, for $\Delta t$ values ranging from 1 to 4, both the direct and total effects meet the robustness criteria in terms of coverage probabilities. However, as we move beyond $\Delta t$ values greater than 4 but less than 28, the delta method encounters coverage issues, particularly affecting the direct effect. The confidence intervals become inflated for the direct effect, resulting in coverage probabilities equal to one. In contrast, MC maintains robust coverage for the indirect and total effects across various $\Delta t$ values. Even at $\Delta t$ values exceeding 28, MC remains reliable for both indirect and total effects. However, coverage drops significantly for the direct effect at $\Delta t$ exceeding 28. Overall, MC consistently performs well, except at $\Delta t$ values greater than 28 for the direct effect. The delta method, while effective at lower intervals, exhibits less robustness at higher intervals. In summary, MC is a better choice for confidence interval coverage, especially when dealing with larger $\Delta t$ values.

### *Statistical Power*

As $\Delta t$ increases, the absolute effect size becomes smaller, approaching zero. Consequently, we expect power to decrease. Except for total effect for $\Delta t$ of 2, power is greater than .8 for $\Delta t$ values ranging from 1 to 4. At $\Delta t = 8$, the delta method's power drops below 0.8, especially at smaller sample sizes. Power starts to drop at a bigger $\Delta t$ of 12 for MC. In summary, while both approaches experience decreasing power as $\Delta t$ increases, the delta method's decline is earlier and faster compared to MC. Considering all factors, MC emerges as a more robust choice for constructing confidence intervals, particularly when dealing with statistical power.

### *Type I Error Rate*

For the delta method, for $\Delta t$ values ranging from 1 to 6, the Type I error rates for both direct and total effects remain within the robustness criteria. However, as $\Delta t$ increases beyond 6, the delta method's Type I error rates get deflated eventually reaching zero. MC exhibits slightly different behavior. While some cases show Type I error rates outside the robustness criteria, especially for higher $\Delta t$ values, the rates remain close to the criteria. Notably, MC maintains robustness even as $\Delta t$ increases. Interestingly, the method behave differently for the indirect effect compared to the direct and total effects. The delta method consistently yields a deflated Type I error rate of zero for all cases. In contrast, MC's Type I error rates, while not entirely deflated, do not meet the robustness criteria. Prior research by Cheung (2009a) and Pesigan and Cheung (2023) has observed similar patterns. Specifically, when both coefficients $\alpha$ and $\beta$ are zero in a simple mediation model, the resulting Type I error rate tends to be deflated. Considering all factors, MC emerges as a more robust choice for constructing confidence intervals, particularly when dealing with Type I errors.

Add a paragraph of conclusions; Move some of the figures to supplementary materials.

## Empirical Example

In this section, we demonstrate the application of a CT mediation model using empirical experience sampling data from an individual with a history of major depressive disorder. Specifically, we estimated the CT-VAR drift matrix and the sampling covariance matrix. These estimates allowed us to generate confidence intervals using both the delta method and MC. Our focus was on examining direct, indirect, and total effects within a model of associations related to depression symptoms.

From a biopsychosocial perspective (inspired by Engel, 1977), we recognize that biology, psychology, and socio-environmental factors are interconnected. Depression, as a mental disorder, manifests with both somatic and emotional symptoms. Somatic symptoms often include physical discomfort and pain. Research has consistently shown that the relationship between depression and pain is bidirectional: depression predicts the development of chronic pain, and chronic pain increases the risk of developing depression (Von Korff & Simon, 1996). Building on this theoretical framework, we investigated the indirect effect of physical discomfort on self-esteem. We hypothesized that this indirect pathway would be mediated by negative affect.

Cite more work on chronic pain theories. Perhaps you can already think about having a more targeted hypothesis based on these theories in terms of the return of self-esteem to its home base.

### Data

In this example, we explore an empirical data set utilized by Ryan and Hamaker (2021), which originates from Wichers et al. (2016) and is further described in Kossakowski et al. (2017). The data set comprises $1,476$ semi-random measurement occasions spanning 239 days. These measurements were collected from a 57-year-old male participant with a history of major depressive disorder. The participant responded to queries multiple times

a day. The time intervals between measurement occasions vary, with a minimum of 13.5 minutes and a maximum of 42.1 hours. We focused on the following variables: physical discomfort, negative affect, and self-esteem. For for physical discomfort, we created a composite score for the following items: "I am hungry", "I am tired", "I am in pain", "I feel dizzy", "I have a dry mouth", "I feel nauseous", "I have a headache", and "I am sleepy". For negative affect, we created a composite score for the following items: "I feel restless", "I feel agitated", "I worry", and "I can concentrate well" (reversed). Finally, self-esteem, we created a composite score for the following items: "I like myself", "I am ashamed of myself" (reversed), "I doubt myself" (reversed), "I can handle anything". All items were measured using a 7-point Likert scale, and the means of these items were calculated to create the composite scores.

### Data Preparation

Before delving into the analysis, we followed a specific data preparation procedure, building upon the approach outlined by Ryan and Hamaker (2021). Here are the steps we took: We created a vector of time values by considering the smallest and largest observed values (0 and 5724.554, respectively). These time values were incremented by 0.10. The newly generated time vector was appended to the existing empirical time vector. We sorted the combined time vector and any duplicated time values were removed. If a time value from the new time vector existed in the original data set, we labeled the observations accordingly. For time values not present in the original data set, we inserted missing values.

### Data Analysis

We fitted the CT-VAR model using the `dynr` package. We estimated the drift $\boldsymbol{\Phi}$ and diffusion $\boldsymbol{\Sigma}$ matrices. Since the variables were scaled prior to model fitting, we assumed that the centralizing tendency $\boldsymbol{\mu}$ is a a vector of zeros. To simplify the model, we did not estimate the measurement model.

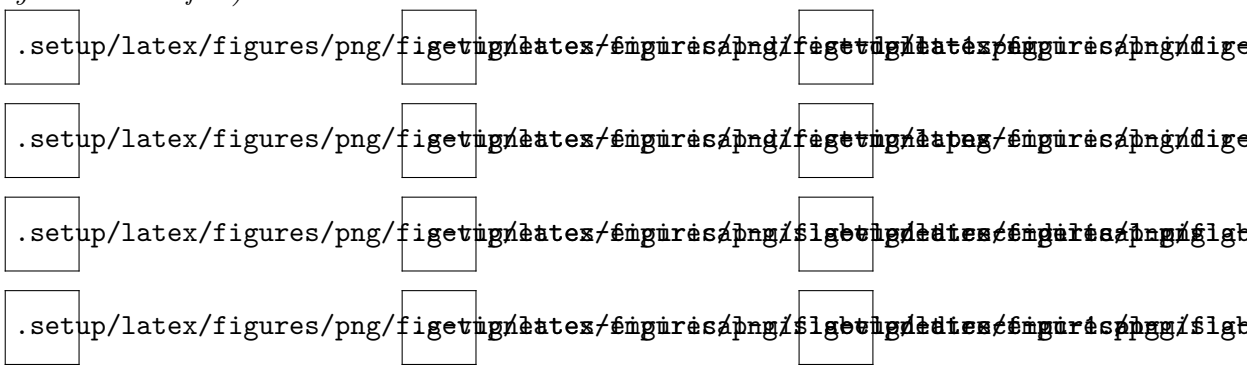
Leveraging the estimated drift matrix $\boldsymbol{\Phi}$ and the corresponding sampling variance-covariance matrix, we calculated direct, indirect, and total effects of the model where the association between self-esteem and physical discomfort is mediated by positive affect using the `cTMed` package. We examined the mediated model across a range of time intervals (from $> 0$ to 15). Confidence intervals were estimated using the delta and MC methods.

### Results

The estimated CT-VAR parameters are presented in Table 4. The direct and total effects were not significant for the time intervals we investigated. The indirect effect was significant for time intervals greater than .6 less than 3.4 for the delta method and for less than 8.70 for MC. The results are presented in the first two rows of Figure 5. Furthermore, we plotted the trajectories of the variables without process noise in Figure 6. We compared the trajectories using the estimated drift matrix $\boldsymbol{\Phi}$ (labeled as "Total") and a modified drift matrix (labeled as "Direct") . We modified the drift matrix by setting coefficients that originate from and go to the mediator (negative effect) to zero except for the autoeffect. To facilitate interpretation, we set the non-significant coefficients to zero. We set the initial states of the variables at high for both physical discomfort and negative affect and low

**Figure 5**

*Regions of Significance for the Direct, Indirect, and Total Effects (Self-Esteem → Negative Affect → Physical Discomfort)*




*Note:* The first two rows correspond to the model where the time intervals are correctly labeled. The last two row correspond to the model where the time intervals are incorrectly label.

self-esteem to show the effect of the coupling of physical discomfort and negative affect on how self-esteem returns to the mean. Notice that for the trajectory of self-esteem on the third row, when the indirect effect is taken out ("Direct"), self-esteem reaches the mean quicker. This suggests that high physical discomfort leads to slower return to the mean for self-esteem when negative affect is considered.

Results showed that the association between physical discomfort and self-esteem is mediated by negative affect. It is notable that both direct and total effects were not significant for the time intervals considered suggesting the importance of the mediator variables in the mechanism linking the somatic distress to an individual's overall evaluation of their own worth. The trajectory plot has also shown that in a scenario where both physical discomfort and negative affect are high, it takes longer for self-esteem to return to the mean when negative affect is considered showing negative affect's importance in the association between physical discomfort and self-esteem.

Can you relate these results to what the theories would predict.

### Mislabeled Time Intervals

We fitted a model where the time intervals are mislabeled. In this model, instead of considering the actual time observations were made, we assumed that the measurement $1,476$ measurement occasions are equally spaced. As such, we recoded the time variable from 0 to $1,475$. This means that the time intervals between observations that actually varied from 13.5 minutes to 42.1 hours in the empirical data set were assumed to be equally spaced with increments of one for the mislabelled case. We used the same data preparation and model specification as the correctly specified time intervals. The CT-VAR drift matrix for this mislabeled model is presented in Table 5. The regions of significance for the direct, indirect, and total effects are presented in the last two rows of Figure 5. The effects are stronger for the mislabeled case. All time intervals have significant results for MC. For the delta method, the indirect and total effects are significant for all time intervals considered.

**Table 4**

*Point Estimates and 95% Confidence Intervals of the CT-VAR Drift Matrix $\boldsymbol{\Phi}$ and the DT-VAR ($\Delta t = 1$) Lagged Coefficients Matrix $\boldsymbol{\beta}$ for the Empirical Data Set from Wichers et al. (2016)*

| Parameter | Estimate | 2.5% | 97.5% | Estimate | 2.5% | 97.5% | 2.5% | 97.5% |
|---|---|---|---|---|---|---|---|---|
| | $CT-VAR$ | | | DT-VAR | | | | |
| | | | | | Delta | | MC | |
| Physical discomfort to physical discomfort ($X \to X$) | −0.898 | −1.057 | −0.740 | 0.406 | 0.342 | 0.470 | 0.346 | 0.474 |
| Physical discomfort to negative affect ($X \to M$) | −0.155 | −0.267 | −0.043 | −0.071 | −0.124 | −0.018 | −0.123 | −0.017 |
| Physical discomfort to self-esteem ($X \to Y$) | 0.005 | −0.124 | 0.135 | 0.011 | −0.047 | 0.069 | −0.047 | 0.069 |
| Negative affect to physical discomfort ($M \to X$) | 0.041 | −0.215 | 0.297 | 0.021 | −0.083 | 0.125 | −0.085 | 0.129 |
| Negative affect to negative affect ($M \to M$) | −0.682 | −0.861 | −0.504 | 0.518 | 0.434 | 0.603 | 0.437 | 0.608 |
| Negative affect to self-esteem ($M \to Y$) | −0.256 | −0.460 | −0.052 | −0.121 | −0.213 | −0.030 | −0.212 | −0.026 |
| Self-esteem to physical discomfort ($Y \to X$) | −0.030 | −0.317 | 0.256 | −0.015 | −0.126 | 0.096 | −0.126 | 0.099 |
| Self-esteem to negative affect ($Y \to M$) | −0.231 | −0.428 | −0.034 | −0.108 | −0.197 | −0.019 | −0.196 | −0.017 |
| Self-esteem to self-esteem ($Y \to Y$) | −0.830 | −1.062 | −0.598 | 0.450 | 0.350 | 0.549 | 0.355 | 0.555 |

*Note.* Estimates for the CT-VAR are calculated using the `dynr` package. Estimates for the DT-VAR are calculated using the `DeltaBeta` (delta method) and `MCBeta` (MC) functions from the `cTMed` package.

**Figure 6**

*Trajectory of Physical Discomfort, Negative Affect, and Self-Esteem Over Time*

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.setup/latex/figures/png/fig-vignettes-empirical-plot-tra

.setup/latex/figures/png/fig-vignettes-empirical-plot-tra

*Note:* "Total" refers to the trajectory using the estimated drift matrix $\mathbf{\Phi}$. "Direct" refers to the trajectory using the estimated drift matrix $\mathbf{\Phi}$ without the effect coming from and going to negative affect except the autoeffects.

The direct effect is significant for time intervals less than 2.83. The mislabeling here resulted in shorter gaps between observations. This pushed the curves to the left and exaggerated the peak especially for the direct and total effects. Compared to the correctly labeled case, in the mislabeled case, it seems that the effects unfold quicker and stronger across time. This exaggerated both the size of the effects and their unfolding across time interval values. The implications for substantive research can be bad. For example, based on the results for the mislabeled case, we can erroneously say that physical discomfort has significant direct, indirect, and total effects on self-esteem, whereas, in the correctly specified model, the effect of physical discomfort on self-esteem is only indirect through negative affect. In the mislabeled case, we can also erroneously assume that the indirect effect peaks faster and diminishes faster compared to the correctly labeled case.

## Discussion

Taking inspiration from the mediation literature, we have extended the work of Deboeck and Preacher (2015) and Ryan and Hamaker (2021) by proposing the delta and MC methods as ways to quantify the uncertainty in the point estimates of direct, indirect, and total effects in CT mediation. We have shown how we build on and extend previous work through using three illustrative examples. Furthermore, we discussed the implications of our contributions to substantive research. We have shown through the simulation study that MC confidence interval is superior to the delta method in terms of coverage, power, and Type I error. Although, when the time interval get large and the population effects are at the boundary of zero, MC starts to have diminished performance. We have shown through the empirical example that the delta and MC methods would have some divergence in conclusion showing that the choice of method has implications on the substantive interpretation of the results. We recommend the use of the delta method as a quick way of exploring regions of significance and then using MC as a way to fine-tune the results. We have also provided a

**Table 5**

*Point Estimates and 95% Confidence Intervals of the CT-VAR Drift Matrix $\boldsymbol{\Phi}$ and the DT-VAR ($\Delta t = 1$) Lagged Coefficients Matrix $\boldsymbol{\beta}$ for the Empirical Data Set from Wichers et al. (2016) with mislabeled time intervals*

| Parameter | Estimate | 2.5% | 97.5% | Estimate | 2.5% | 97.5% | 2.5% | 97.5% |
|---|---|---|---|---|---|---|---|---|
| | $CT-VAR$ | | | DT-VAR | | | | |
| | | | | | Delta | | MC | |
| Physical discomfort to physical discomfort ($X \to X$) | −1.570 | −1.820 | −1.319 | 0.206 | 0.152 | 0.260 | 0.157 | 0.266 |
| Physical discomfort to negative affect ($X \to M$) | −0.326 | −0.492 | −0.161 | −0.101 | −0.150 | −0.052 | −0.149 | −0.051 |
| Physical discomfort to self-esteem ($X \to Y$) | 0.195 | 0.019 | 0.370 | 0.068 | 0.018 | 0.118 | 0.017 | 0.117 |
| Negative affect to physical discomfort ($M \to X$) | 0.041 | −0.315 | 0.397 | 0.012 | −0.071 | 0.095 | −0.073 | 0.099 |
| Negative affect to negative affect ($M \to M$) | −1.067 | −1.305 | −0.829 | 0.365 | 0.289 | 0.441 | 0.291 | 0.446 |
| Negative affect to self-esteem ($M \to Y$) | −0.339 | −0.590 | −0.088 | −0.114 | −0.191 | −0.036 | −0.192 | −0.034 |
| Self-esteem to physical discomfort ($Y \to X$) | −0.014 | −0.397 | 0.370 | −0.006 | −0.091 | 0.080 | −0.091 | 0.083 |
| Self-esteem to negative affect ($Y \to M$) | −0.395 | −0.651 | −0.140 | −0.133 | −0.212 | −0.055 | −0.211 | −0.053 |
| Self-esteem to self-esteem ($Y \to Y$) | −1.136 | −1.405 | −0.866 | 0.343 | 0.263 | 0.422 | 0.266 | 0.427 |

*Note.* Estimates for the CT-VAR are calculated using the `dynr` package. Estimates for the DT-VAR are calculated using the `DeltaBeta` (delta method) and `MCBeta` (MC) functions from the `cTMed` package.

free and open-source `R` package `cTMed` to implement the methods we have proposed together with miscellaneous functions to make the interpretation and presentation of the results easier. Of particular importance is the plotting capabilities of the package which facilitates visualizing how effects change overtime and regions of statistical significance with respect to the time interval. The package will be a valuable tool for applied researchers in using the methods we proposed in their substantive area of research.

We note some data analytic decisions we took to address some issues. In the illustrative examples and simulation study, we estimated the measurement model to account for measurement error. We used single indicators for each latent variables with factor loadings fixed to one and residual variances that are freely estimated. While it is common especially in DT-VAR to ignore the measurement model, doing so has consequences for the estimates of parameters in the dynamic structure (Schuurman & Hamaker, 2019; Schuurman et al., 2015; Staudenmayer & Buonaccorsi, 2005). Another decision we made relates to data techniques for handling irregularly spaced data. Within the CT framework as irregularly spaced data can be treated as a missing data problem, one technique that can be used to facilitate fitting CT models is by choosing a time scale that results in a small enough time interval $\Delta t$ and inserting missing data in between observed cases. This approach, referred to as oversampling (Singer, 2012; Voelkle & Oud, 2012), reduce the time intervals between successive observations helping improve convergence. This aided in the optimization process but lengthened the time needed as insertion on many missing values resulted in a longer time series, extending the time needed for convergence. Note that despite this solution other numerical issues are still present. In `dynr`, the SDEs are solved via numerical integration as opposed to using direct analytic solutions. Doing so allows for the possibility of fitting nonlinear SDEs, but this approach could also create more numerical problems in the estimation and Hessian computation process.

### Limitations and Future Directions

We acknowledge some of the limitations of our work and propose future directions. The current study employed a frequentist approach for modeling CT-VAR using the `dynr` package. In the current approach, the parameters are fixed and do not vary across individuals. A multilevel mediation perspective where the repeated measures are nested within an individual (Kenny et al., 1998) is a logical extension of CT mediation. In this approach, coefficients for each individual have fixed and random components. Work on Bayesian mediation model (e.g., Enders et al., 2013; Y. Yuan & MacKinnon, 2009) makes multilevel mediation easier compared to the frequentist approach. A Bayesian hierarchical or multilevel CT-VAR model is a more flexible alternative to the current approach (Driver & Voelkle, 2018; Hecht & Zitzmann, 2020; Oravecz et al., 2011, 2018). In the Bayesian CT-VAR framework, distributions of the direct, indirect, and total effects can be generated by calculating them from the posterior distribution of the drift matrix $\boldsymbol{\Phi}$. Credible intervals using the appropriate percentiles can be obtained from these distributions. To implement this approach, we can use the `PosteriorMed` function in the `cTMed` package. Alternatively, we can treat the posterior mean vector and covariance matrix of the posterior distribution of the drift matrix $\boldsymbol{\Phi}$ as frequentist estimates which can then be used in the delta and MC methods. Further exploration and investigation of these ideas are warranted to better understand their implications and performance.

The current study did not explore the use of bootstrapping due to its computational expense. However, we recognize that it could potentially be applied in our current context. Here are two approaches to consider. Nonparametric bootstrapping where we resample the data by individual with replacement. Essentially, it is similar to typical nonparametric bootstrapping with a wide data set. Parametric bootstrapping where we generate raw data from the estimated CT-VAR model. In both cases, the CT-VAR model needs to be fitted on each bootstrap sample. However, this multiple model fitting becomes the bottleneck in the process. Looking ahead, a promising area for future research involves exploring optimization procedures that can make multiple model fittings less computationally expensive. This would allow bootstrapping to be a viable option for VAR models. For the time being, the delta and MC methods can be used in place of the bootstrap.

In our study, we have considered a perspective where all variables in the system are associated with each other. While this approach works well for small models, it becomes problematic for large, high-dimensional systems. The reason lies in the potential explosion of parameters when all variables are interconnected. To address this issue, we recommend using regularization techniques, specifically the lasso (Least Absolute Shrinkage and Selection Operator) introduced by Tibshirani (1996, 2011). The lasso method effectively shrinks parameters, introducing sparsity in the drift matrix $\mathbf{\Phi}$. The implications of this regularization approach extend to calculating direct, indirect, and total effects in high-dimensional systems. Notably, parameters that are reduced to zero by lasso can be treated as fixed when applying the delta and MC methods. However, to validate the performance of these adjustments in practice, further investigation is needed.

In our current work, we adopted a path tracing perspective, focusing on understanding the temporal relationships between variables. However, we did not explore the network analysis perspective. In contrast, Ryan and Hamaker (2021) take a network analysis approach when studying CT mediation models. They propose two centrality measures: total effect centrality and indirect effect centrality. Total effect centrality summarizes the total effect of a variable on all other variables within the network over a specific time interval. It quantifies the cumulative impact of an impulse intervention on a variable, affecting the entire system. Essentially, it reveals the overall shock to the network. Researchers can use total effect centrality to identify optimal targets for pulse interventions that can lead to significant system-wide changes. By examining how the cumulative effect evolves over time following the intervention, researchers can determine the most effective timing for interventions. Indirect effect centrality focuses on the role of a specific variable as a mediator of relationships between other variables in the network. It captures the change in the effect of impulses on one variable due to press interventions on a mediator variable, keeping it fixed at all time points. This measure helps identify variables that play a crucial role in mediating relationships between other variables in the network. By focusing on indirect effects mediated by a specific variable, researchers can pinpoint key targets for press interventions that influence how other variables interact with each other. These centrality measures provide valuable insights into the dynamics of network systems. They guide researchers in identifying effective intervention targets for bringing about desired changes in the network structure and dynamics. Additionally, both total and indirect effect centrality measures, along with the delta and MC method confidence intervals, are implemented in the `cTMed` package. However, further investigation is needed to assess the finite sample performance

of these methods.

In the empirical example where data is collected at different moments within the day, there is often a gap of missing observations between the last measurement for the day and the first measurement for the next day. This is known as the night gap problem in intensive longitudinal data. We treated this gap as missing data and modeled the dynamics in a single level. However, doing so assumes that the dynamic process continues during this gap although we are not able to observe it. This is an assumption that may be tenable for some variables but not for others. For example, it might not be a reasonable assumption to say that the dynamics of physical discomfort, negative affect, and self-esteem is the same when the individual is awake and when the individual is sleeping. Future studies can take a multilevel approach where observations are nested within the day to account for this night gap problem.

Lastly, we relied on data that follows a normal distribution. However, in substantive research, data can take various forms, such as categorical data (e.g., gender and race), ordinal data (e.g., educational level), count data (e.g., the number of drinks consumed in an episode). Given this diversity of data types, it is essential to explore suitable estimation methods suitable when dealing with nonnormal data. Additionally, we should investigate the implications of using nonnormal data in CT-VAR mediation models.

### Concluding Remarks

Overall, the paper contributes to advancing the methodology for estimating direct, indirect, and total effects in CT mediation models and provides valuable insights for researchers in the field of longitudinal data analysis and mediation modeling. Additionally, it provides a free and open-source software package `cTMed` to implement the methods described.

## Open Practices Statement

The data and materials for this study are available on OSF (https://osf.io/qwnmf/?view_only=855be2012b554f05bbc679946845bab8) and GitHub (https://github.com/jeksterslab/manCTMed, https://jeksterslab.github.io/manCTMed/index.html). Source code and documentation for the `cTMed` `R` package are available on GitHub (https://github.com/jeksterslab/cTMed, https://jeksterslab.github.io/cTMed/index.html).

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